Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Differentiation

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Differentiate $y = x^4$. **[2]**

2. Differentiate $y = 3x^5 - 2x^2 + 7$. **[3]**

3. Differentiate $y = \dfrac{1}{x}$. **[2]**

4. Differentiate $y = \sqrt{x}$. **[2]**

5. Differentiate $y = x^{3/2}$. **[2]**

6. Differentiate $y = 2x^{-3} + 5x$. **[3]**

7. Find $\dfrac{dy}{dx}$ if $y = (2x + 1)^3$ (chain or expand). **[4]**

8. Find the gradient of $y = x^3 - 3x$ at $x = 2$. **[3]**

9. Find the equation of the tangent to $y = x^2 + 1$ at $x = 2$. **[4]**

10. Find the equation of the tangent to $y = x^3 - x$ at $x = 1$. **[4]**

11. Find the equation of the normal to $y = x^2 - 4x$ at $x = 3$. **[4]**

12. Find the stationary points of $y = x^3 - 3x + 2$. **[5]**

13. Find the stationary points of $y = x^2 - 6x + 5$. **[4]**

14. Determine the nature of the stationary points of $y = x^3 - 12x$. **[5]**

15. Differentiate $y = \dfrac{x^2 + 1}{x}$. **[3]**

Section B – Hard (Q16–25)

16. Find the equation of the tangent to $y = x^3 - 3x^2 + 2$ at the point where $x = 1$. **[5]**

17. Find the coordinates of the stationary points of $y = 2x^3 - 9x^2 + 12x$ and determine their nature. **[7]**

18. A curve has equation $y = x + \dfrac{4}{x}$. Find the coordinates of the stationary points and determine their nature. **[6]**

19. Find the values of x for which the gradient of $y = x^3 - 6x^2 + 5$ is zero. **[4]**

20. The normal to $y = x^2 - 3x$ at $x = 2$ meets the x-axis at P. Find the coordinates of P. **[5]**

21. Differentiate $y = (x^2 + 1)(x - 3)$ using the product rule. **[4]**

22. Differentiate $y = \dfrac{x+1}{x-1}$ using the quotient rule. **[4]**

23. Find $\dfrac{d^2y}{dx^2}$ if $y = x^4 - 3x^2 + x$. **[3]**

24. Show that $y = x^3 - 3x$ has a point of inflection at the origin. **[4]**

25. The curve $y = ax^2 + bx + c$ has a stationary point at $(1, 2)$ and passes through $(0, 3)$. Find a, b, c. **[6]**

End of Worksheet • *Show all working clearly.*